Unit: 6 Unit 8: Quadratics **Name:** ______________

Topic: Converting Forms (Completing the Square) **Date:** __________

Advanced (Habanero)

Q1. What is the vertex form of $x^2 + 6x + 8$?
(A) $(x+3)^2 - 1$
(B) $(x+4)^2 - 8$
(C) $(x+3)^2 + 1$
(D) $(x+2)^2 - 4$
(E) $(x+4)^2 + 8$

Q2. Complete the square for $x^2 + 5x$
(A) $x^2 + 5x + \frac{25}{4}$
(B) $x^2 + 5x - \frac{25}{4}$
(C) $x^2 - 5x + \frac{25}{4}$
(D) $x^2 + 5x + 25$
(E) $x^2 - 5x - 25$

Q3. Which equation represents a parabola with vertex $(2, 3)$?
(A) $y = (x-2)^2 + 3$
(B) $y = (x+2)^2 - 3$
(C) $y = -(x-2)^2 + 3$
(D) $y = -(x+2)^2 - 3$
(E) $y = (x-2)^2 - 3$

Q4. What is the value of c in x^2+4x+c if the vertex is $(-2, 1)$?
(A) -3
(B) -1
(C) 0
(D) 1
(E) 3

Q5. Convert $x^2 - 2x - 3$ to vertex form
(A) $-(x-1)^2 - 4$
(B) $-(x+1)^2 - 4$
(C) $(x-1)^2 - 4$
(D) $(x+1)^2 - 4$
(E) $-(x+1)^2 + 4$

Q6. Complete the square to solve $x^2 + 2x + 1 = 0$
(A) $(x+1)^2 = 0$
(B) $(x-1)^2 = 0$
(C) $x^2 = 0$
(D) $x = 0$
(E) $x = -1$

Q7. Which equation is equivalent to $x^2 + 10x + 20$?
(A) $(x+5)^2 - 5$
(B) $(x+5)^2 + 5$
(C) $(x-5)^2 + 5$
(D) $(x-5)^2 - 5$
(E) $(x+10)^2 + 20$

Q8. What is the vertex of $y = -x^2 - 6x - 5$?
(A) $(-3, 4)$
(B) $(-3, -4)$
(C) $(3, -4)$
(D) $(3, 4)$
(E) $(-3, 5)$

Open Ended Questions (FRQ)

Q9. Convert the equation $x^2 + 8x + 12 = 0$ to vertex form and find the vertex

Q10. Complete the square to solve the equation $x^2 + 4x + 3 = 0$. Show all steps

Chef's Tips
- Use exact values for calculations
- Show all work for free response questions
- Check for extraneous solutions in free response questions